P#7 - Rapid Prototyping

Score Companion

A rehearsal score companion for capturing corrections quickly and turning them into focused individual practice.

Team 08 | CMPS 453 | October 12, 2026

Role	Code	Name
Team Lead / Integration	A	Matthew Pisani
Screen Design	B	Logan Broussard
Screen Design	C	Ankit Joshi
Screen Design	D	Jonathan Dempsey

Static prototype: six integrated screen states. No working application is claimed.

Delivery Schedule

Expected dates below come from the October 11 GitLab tracker snapshot. Delivered dates use recorded issue-close timestamps converted to America/Chicago. Later closures are unverified; receipt of a design is distinguished from issue closure. No dates have been backdated.

Component / issue	Owner	Expected	Delivered / verified status
GitLab setup	A	Before work	Milestone created Oct 7; no separate setup issue
Segments #9	Team / A	Oct 7	Oct 7 (recorded closure)
Critical moment #10	Team / A	Oct 8	Oct 7 (recorded closure)
MVP features #11	Team / A	Oct 10	Oct 7 (recorded closure)
Screen design #12	B	Oct 10	Oct 10 (recorded closure)
Screen design #13	C	Oct 10	Oct 11 (recorded closure)
Screen design #14	D	Oct 11	Design received Oct 12; closure unverified
Screen integration #15	A	Oct 10	Prepared Oct 12; closure unverified
README #16	A	Oct 12	Prepared Oct 12; closure unverified
Final submission #16	A	Oct 12	Pending Moodle submission

Repository: https://gitlab.com/ull-gitlab/453/fa26/fa26-team-08/music-project

Milestone: P#7 - Rapid Prototyping. Issue #16 combines README and final submission. The integration due date predates some screen-design due dates; the dates above retain the actual tracker record.

Part 1: Behavioral Segments

Segment 1 - Preparation-First Internalizers

Description: These musicians reduce dependence on the score by preparing difficult material before rehearsal, retaining corrections mentally, and practicing until responses become automatic.

Key Behaviors:

Prepare difficult passages before rehearsal and practice them repeatedly.

Use mental recall and muscle memory to apply corrections.

Pain Points:

Unfamiliar or complex material must sometimes be performed before there is enough time to internalize it.

Several rapid corrections can exceed what they can retain and practice between run-throughs.

Evidence:

“No struggle, I memorize the music prior.” - Interview 2

“Yeah. I remember it, and then I practice it the right way until it becomes muscle memory.” - Interview 1

Success looks like: Corrections become reliable responses through targeted preparation and practice.

Part 1: Behavioral Segments

Segment 2 - Reactive Score Annotators

Description: These musicians use the score as an external memory by adding measure numbers, transition cues, rhythm reminders, accidentals, and other context-specific marks during or after rehearsal.

Key Behaviors:

Mark important measures, transitions, rests, and performance problems on the score.

Use a quick temporary mark during rehearsal, then return later to add or interpret a useful cue.

Pain Points:

Rehearsal can move too quickly to write a complete note without jeopardizing the next entrance.

Too many notes, or a mark without context, can clutter the score and weaken later recall.

Evidence:

“I write the measure numbers on important spots and transitions. I also write ear-mapping signals over long multi-measure rests.” - Interview 7

“Sometimes, on horn at least I will mark it if something’s flat so I don’t have to go through the mental problem.” - Interview 6

Success looks like: A correction is captured without missing the next entrance and remains understandable later.

Part 1: Behavioral Segments

Segment 3 - Format-Constrained Score Navigators

Description: These musicians spend rehearsal effort accessing, moving through, or visually reading the score. They need reliable access, legible notation, and navigation that fits the performance context.

Key Behaviors:

Download score files for reliable access and search for the assigned part when needed.

Zoom, scroll, turn pages, or adjust the display while playing.

Pain Points:

A score may not open when connectivity is unreliable, and shared storage can slow access to the correct part.

Cramped or obstructed displays and page movement can interrupt performance.

Evidence:

“If you don't download it and you try to open it at the football games, you're not getting it. It's not happening.” - Interview 1

“MuseScore doesn't have a good zoom functionality to it and the UI gets in the way of the notes. No markup either for finger positions.” - Interview 4

Success looks like: The correct assigned part opens offline and the needed passage stays readable and easy to reach.

Part 2: Critical Moment

The Moment

During rehearsal, a musician receives a correction for a difficult measure or transition and must be ready to play again almost immediately. The product is most valuable when the musician needs to preserve the correction now and revisit the exact passage later without missing the next entrance.

Trigger

The director gives feedback about a specific measure or transition, then restarts the passage or moves on before the musician can write a complete note.

Context

The musician is viewing the assigned score during an active rehearsal, where attention must alternate quickly between the conductor, the score, and the next entrance.

User Goal

Capture enough context to return to the corrected passage during individual practice, apply the correction, and perform it confidently at the next run-through.

Current Failure

Paper notes can take too long during a rapid rehearsal cycle. Generic score viewers can make annotation or navigation awkward and can clutter notation, while separate practice tools require the musician to switch tools or rely on memory.

Your Solution

Keep score access, page or section navigation, quick measure markers, concise practice cues, and unresolved-passage review in one focused workflow. The musician can mark the relevant measure quickly, remain ready to play, and later return to a focused list of marked passages for practice.

Evidence

“Often we don’t have enough time to pick up our pencils, write it down, and get ready to play again, because [the director]’s already conducting what he wants us to fix.” - Interview 7

The single critical instant is receiving a correction just before playing resumes. Opening the score and later practice are supporting steps around that moment.

Part 3: MVP Feature List

Exactly five features support the rehearsal-to-practice flow. Each records an interview insight, a behavioral segment, a reduced problem, and a remaining assumption.

Feature 1 - Offline Score Library

Description: Store assigned score files locally in one predictable library so a musician can open the correct part without relying on a connection.

Interview Insight: “If you don't download it and you try to open it at the football games, you're not getting it. It's not happening.” - Interview 1

Segment: Format-Constrained Score Navigators.

Problem Reduced: The musician cannot reliably begin the rehearsal workflow when the assigned score is unavailable or difficult to find in shared storage.

Assumption: Musicians will download and organize their assigned parts before rehearsal or performance.

Feature 2 - Uncluttered Score Viewer with Page and Section Navigation

Description: Display the selected score clearly and provide direct page or section controls for reaching the needed passage.

Interview Insight: “MuseScore doesn't have a good zoom functionality to it and the UI gets in the way of the notes. No markup either for finger positions.” - Interview 4

Segment: Format-Constrained Score Navigators.

Problem Reduced: Small or obstructed notation and awkward navigation can interrupt a musician before or during the corrected passage.

Assumption: Clear page and section controls can reduce navigation time enough to avoid disrupting performance.

Feature 3 - One-Action Measure Marker

Description: Let a musician place a quick marker on the current measure while receiving a rehearsal correction.

Interview Insight: “Often we don’t have enough time to pick up our pencils, write it down, and get ready to play again.” - Interview 7

Segment: Reactive Score Annotators.

Problem Reduced: A complete note takes too much time during a rapid rehearsal cycle, but relying only on memory can lose the correction.

Assumption: A single, fast digital action can capture a useful reminder without causing the musician to miss the next entrance.

Feature 4 - Measure-Linked Practice Cue

Description: Allow the musician to add a short cue to a marked measure after the immediate rehearsal pressure has passed.

Interview Insight: “I write the measure numbers on important spots and transitions. I also write ear-mapping signals over long multi-measure rests.” - Interview 7

Segment: Reactive Score Annotators.

Problem Reduced: A temporary mark alone may not explain what needs to change, while extensive notes can clutter the score.

Assumption: A concise, measure-linked cue gives enough context for later practice without obscuring notation.

Feature 5 - Unresolved-Passage Review

Description: Show a focused list of marked measures and cues so the musician can select a passage, practice it, and mark it resolved.

Interview Insight: “Yeah. I remember it, and then I practice it the right way until it becomes muscle memory.” - Interview 1

Segment: Preparation-First Internalizers and Reactive Score Annotators.

Problem Reduced: Musicians need a focused way to turn rehearsal corrections into targeted individual practice instead of searching through the entire score or relying on memory.

Assumption: Musicians will return to unresolved markers and find the focused list more useful than a fully annotated score or memory alone.

Features Not Included

Feature / proposed idea	Why excluded
Automatic transcription / error detection	Requires interpretation beyond the evidence-backed capture-and-practice workflow.
Social feed / ensemble chat	No interview evidence identifies it as necessary for the critical moment.
Replacing all paper workflows	Some musicians intentionally use paper or mixed workflows.
Tuner, metronome, audio playback	Appeared in D's sketch; outside the agreed five-feature MVP and unnecessary for capturing a correction.
Online browsing, export, login, 2FA, account links and general settings	Appeared in B's sketch; expand the task beyond downloaded score access, marking, cues and review.

Part 4: Integrated Prototype

Task: open the assigned score, capture a rehearsal correction at measure 24 without delaying the next entrance, then preserve context and practice it later.

Flow: Library > Score viewer > Quick marker > Practice cue > Review passage > Resolved confirmation.

Integration decisions and attribution

Matthew (A) redrew the selected ideas in one consistent layout, kept navigation outside the score, retained the same score and measure across states, made the marking target explicit, and added a completion state. These are integrated revisions of the supplied designs, not claims that the original contributors produced these exact final screens.

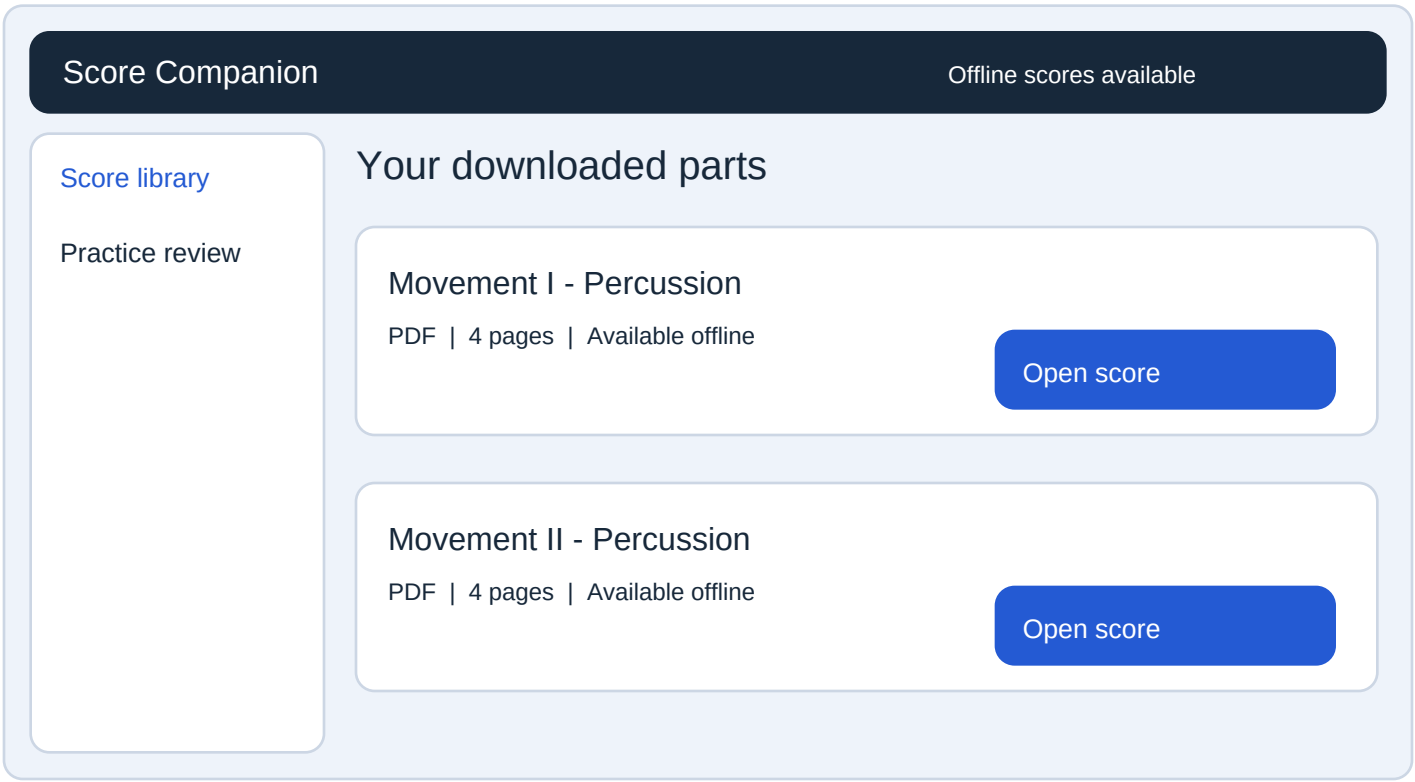
Member	Source contribution retained
B - Logan	Personal score library, large score view, annotation concept.
C - Ankit	Offline status, complete MVP flow, quick marker, deferred cue, review list and evidence annotations.
D - Jonathan	Score library, page indicator and score-view layout.
A - Matthew	Selection, scope reconciliation, common layout, one-tap target, labels, evidence mapping and new resolved screen.

The appendix preserves the original submissions. Controls in those originals that fall outside the MVP remain excluded from the integrated flow.

Sample score titles, notes and cue text are illustrative. The static screens demonstrate proposed behavior; offline storage, PDF loading, marking and saving are not implemented.

Screen 1: Offline score library

B: personal music library; C: offline library and flow; D: library layout. Integrated/redrawn by A.



What the user sees: Downloaded instrument parts with an offline badge and a clear Open score action.

What the user does: Open Movement I - Percussion from the downloaded library.

Design decision: Combine a predictable personal library with clear offline availability. Avoid browse categories and online discovery.

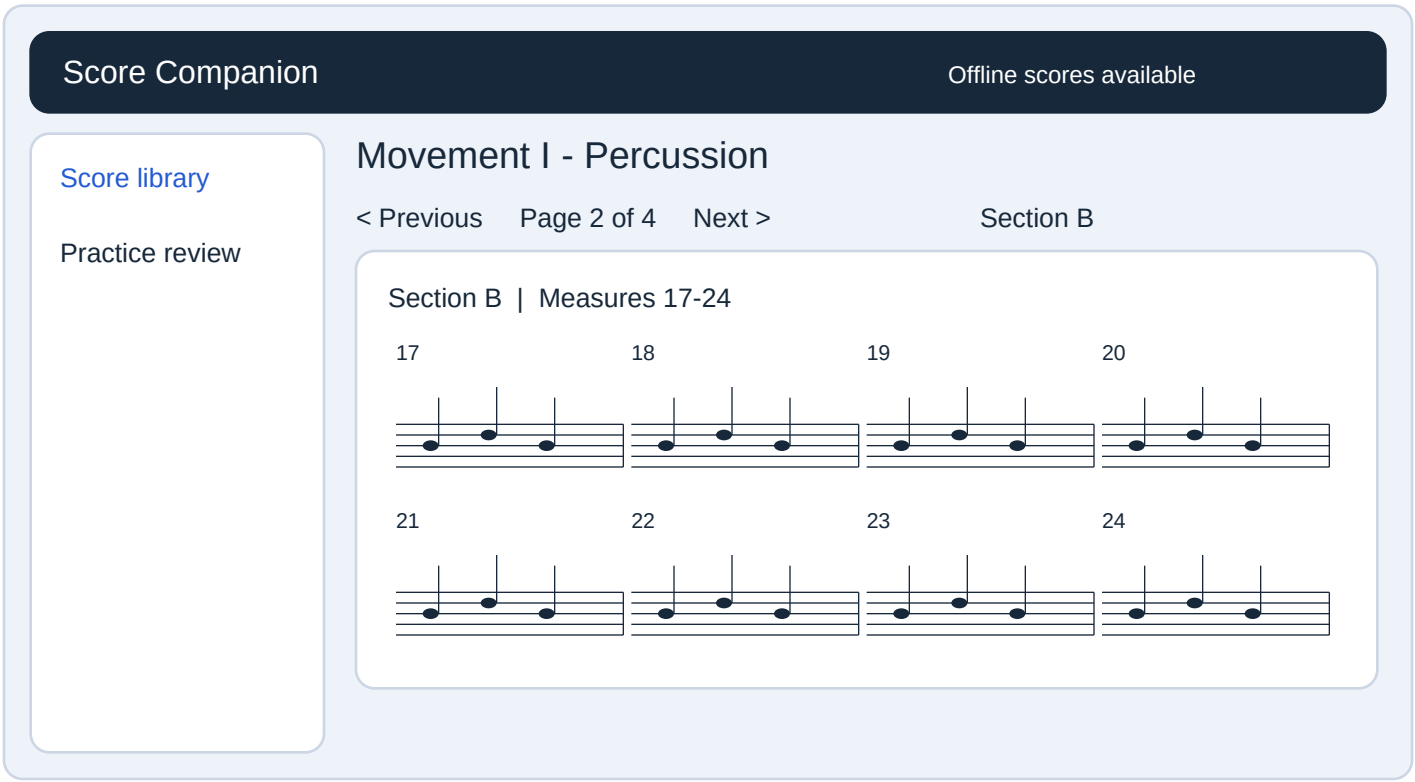
Interview evidence: “I download everything to where it's all in the same place in the offline section so I can easily bounce between.” - Interview 1

Assumption to test: Users download assigned parts before rehearsal.

Next: Score viewer and navigation

Screen 2: Score viewer and navigation

B: notation view; C: navigation outside score; D: page indicator and score viewer. Integrated/redrawn by A.



What the user sees: A readable score with page and section controls outside the notation.

What the user does: Use page or section controls to reach Section B, measure 24.

Design decision: Combine Jonathan's page indicator and Logan's large notation area with Ankit's external navigation controls.

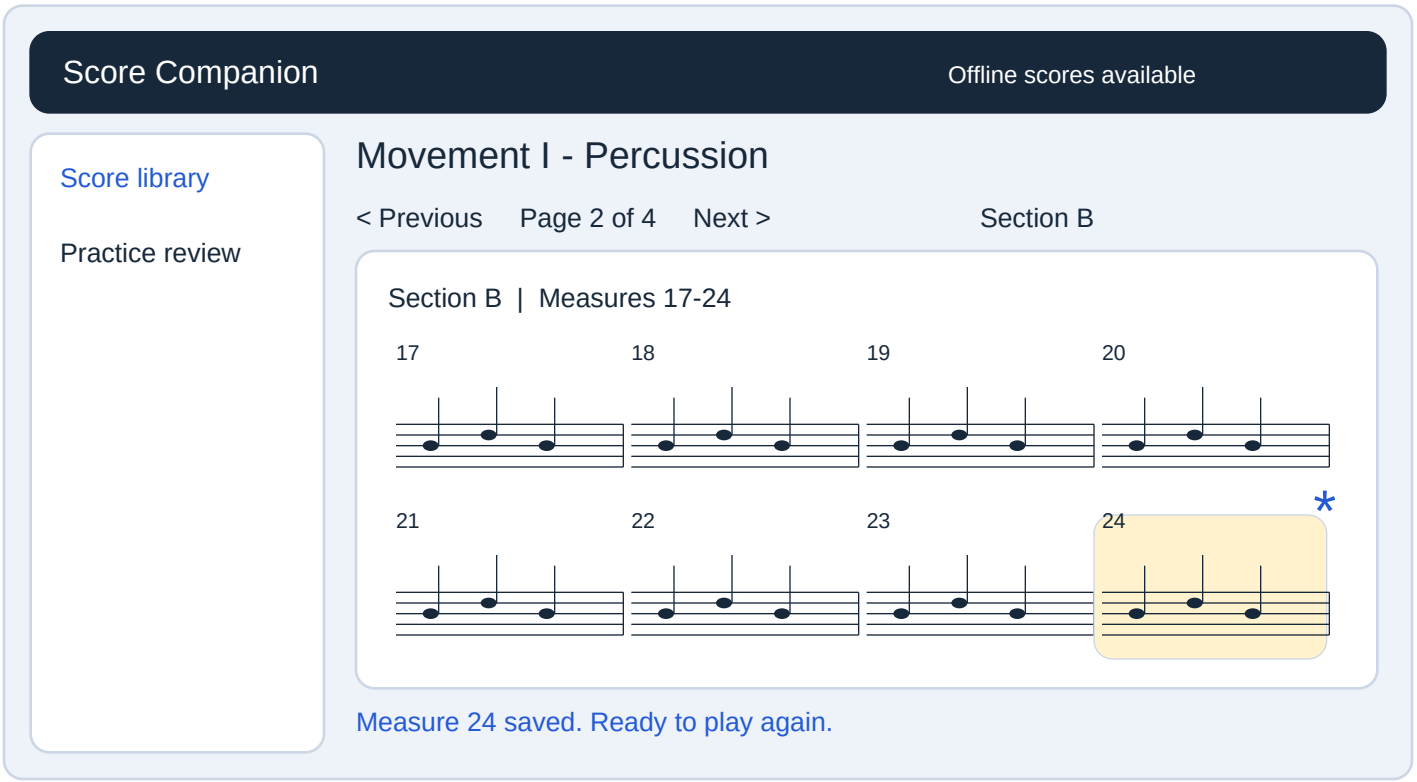
Interview evidence: “MuseScore doesn't have a good zoom functionality to it and the UI gets in the way of the notes. No markup either for finger positions.” - Interview 4

Assumption to test: Navigation is fast enough to avoid interrupting performance.

Next: One-action measure marker

Screen 3: One-action measure marker

C: quick-marker concept and evidence. A: explicit tap target, confirmation, and integrated redraw.



What the user sees: Measure 24 highlighted with a star and a short saved confirmation.

What the user does: Tap measure 24 once when the director calls out the correction; resume playing without typing.

Design decision: Make the measure itself the tap target. One tap stores the marker; confirmation does not block the next entrance.

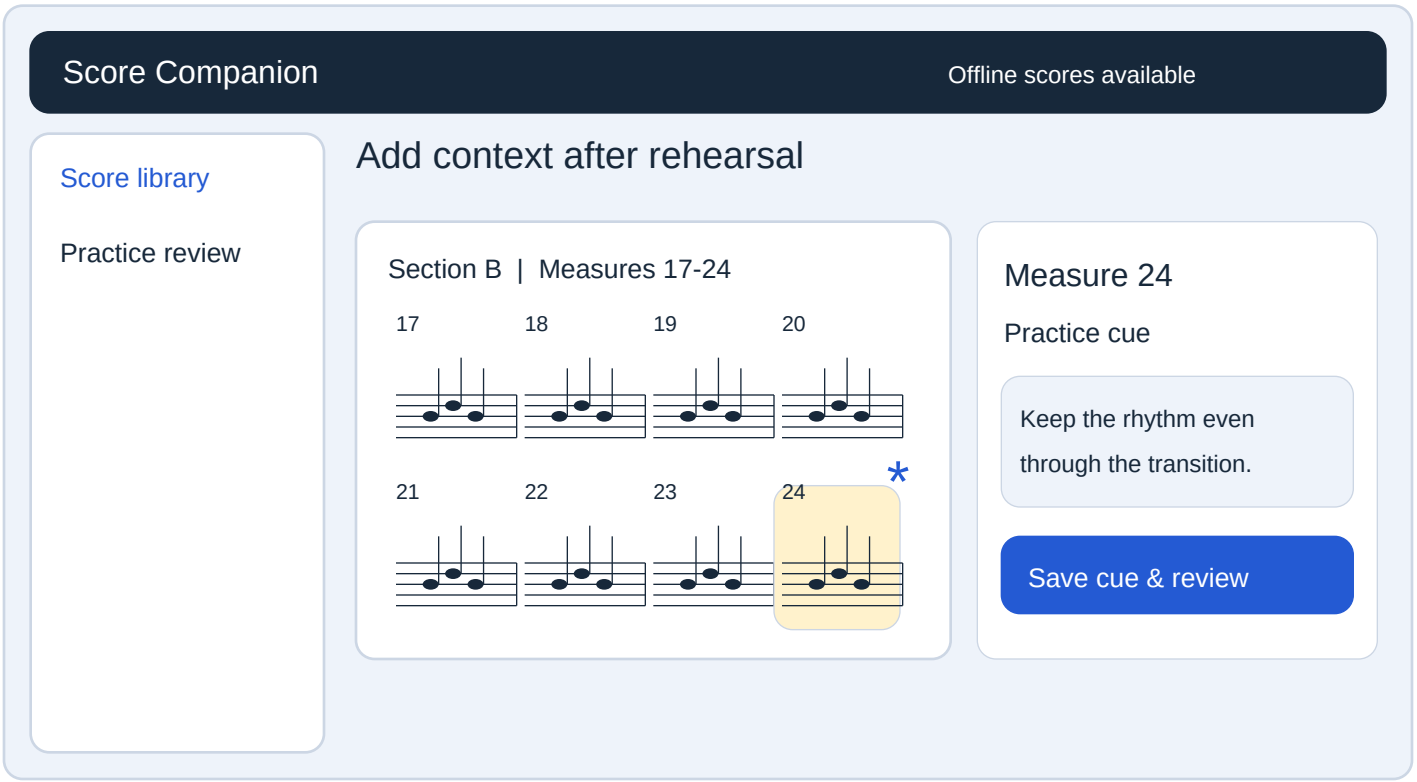
Interview evidence: “Often we don’t have enough time to pick up our pencils, write it down, and get ready to play again.” - Interview 7

Assumption to test: One tap is precise and fast enough to preserve readiness.

Next: Measure-linked practice cue

Screen 4: Measure-linked practice cue

B: annotation concept; C: deferred measure-linked cue. Integrated/redrawn by A.



What the user sees: The marked passage alongside a short cue field.

What the user does: After rehearsal, type a concise reminder and select Save cue & review.

Design decision: Keep Logan's annotation concept but constrain it to a brief measure-linked cue entered after rehearsal.

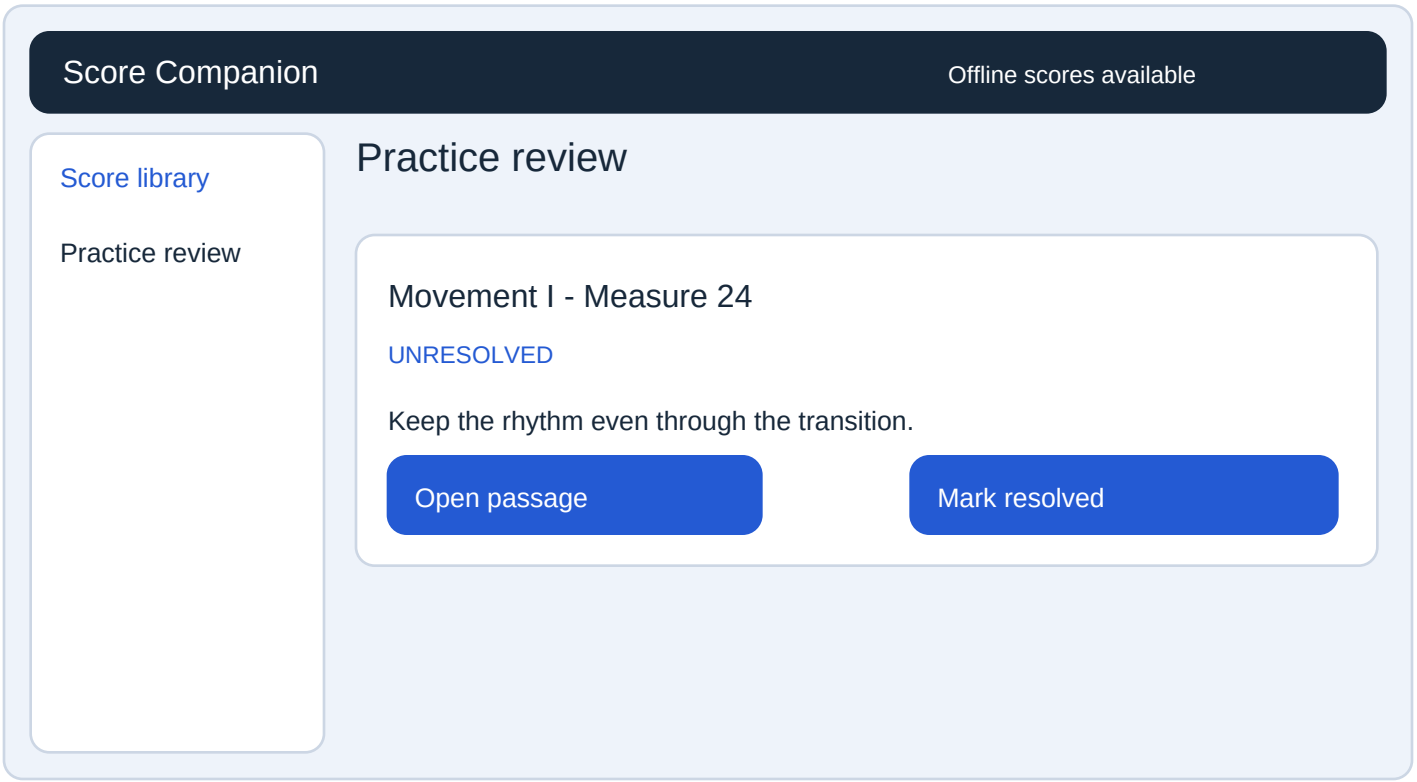
Interview evidence: “I star the measure. And when I look at it later by myself, I’ll see the star and write it in.” - Interview 7

Assumption to test: A short cue preserves enough context for later practice.

Next: Unresolved-passage review

Screen 5: Unresolved-passage review

C: unresolved list, passage access, and resolved action. Integrated/redrawn by A.



What the user sees: A focused list containing measure 24 and its saved correction cue.

What the user does: Select Open passage, practice the correction, then select Mark resolved.

Design decision: Retain Ankit's focused review list and separate practice from the musician's explicit resolved decision.

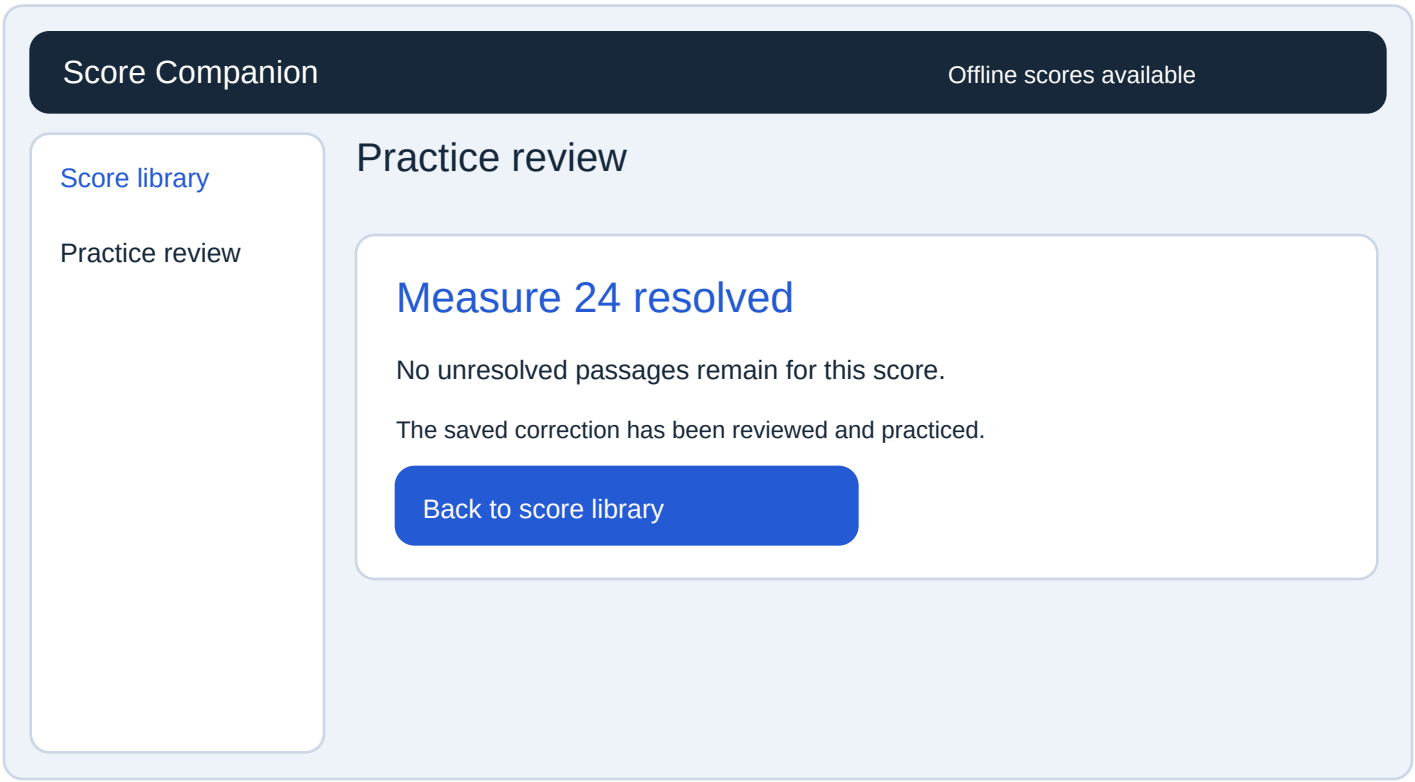
Interview evidence: “Yeah. I remember it, and then I practice it the right way until it becomes muscle memory.” - Interview 1

Assumption to test: Users revisit markers and accurately judge when a correction is resolved.

Next: Correction resolved

Screen 6: Correction resolved

A: new completion screen extending C's resolved action.



What the user sees: A resolved confirmation and an empty unresolved list for this score.

What the user does: Confirm that measure 24 is resolved; return to the score library.

Design decision: Add a visible completion state so the task ends with a clear result rather than only an instruction.

Interview evidence: “I remember it, and then I practice it the right way until it becomes muscle memory.” - Interview 1

Assumption to test: A visible completion state makes the result clear; resolution is self-reported.

Next: Task complete.

Appendix: Tools and Evidence

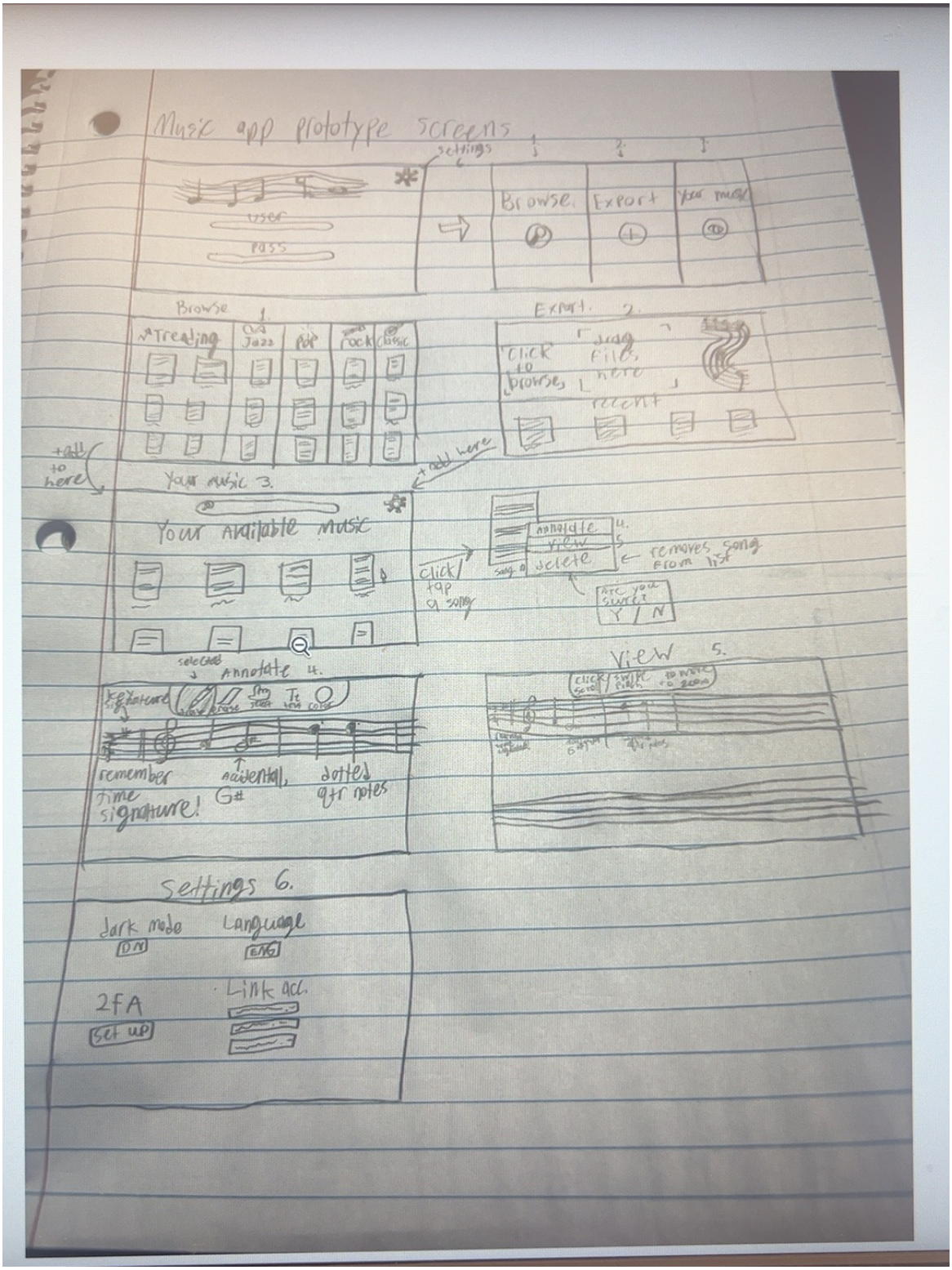
Tool / source	Purpose
Team Markdown task documents	Source for segments, critical moment, MVP justifications and exclusions.
Paper sketches (B and D)	Original library, viewer and annotation concepts.
Ankit's PDF prototype	Source for the five-step MVP workflow and evidence annotations.
ChatGPT / Codex and ReportLab	Assisted integration, final vector mockups, annotations, report and README assembly.
GitLab tracker snapshot	Issue assignments, expected dates and recorded closures; snapshot dated Oct 11.

Interview references identify numbered interviews in p4/complete-combined-interview-scripts-standardized.txt. Short quotes retain the team's anonymized interview numbering; participant names and raw transcripts are not reproduced here.

Validation still needed: observe whether the one-tap marker preserves readiness during a timed rehearsal simulation, whether cues remain understandable later, and whether the review list is used. Static screens do not establish these outcomes.

Original B: Logan Broussard

Source submission. Library/view/annotation concepts informed the integrated flow. Other controls shown are excluded concepts, not additional MVP features.



Original D: Jonathan Dempsey

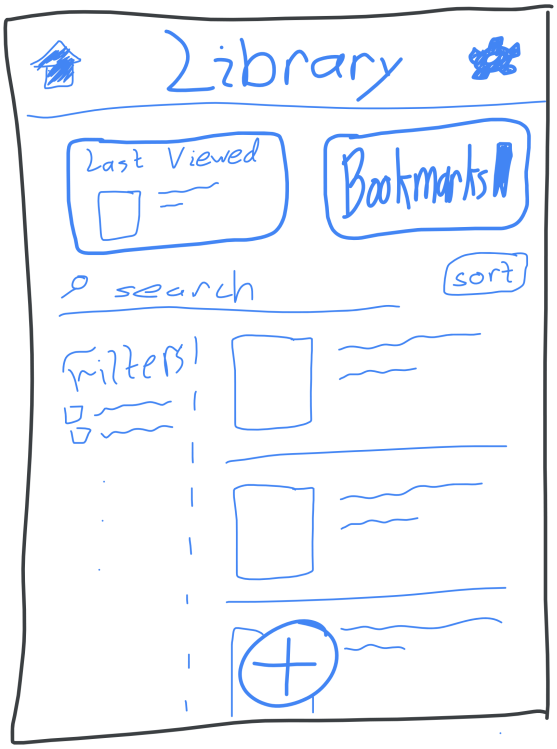
Source submission, home and score viewer. Library, viewer and page indicator informed integration. Tools and unrelated controls remain outside the MVP.

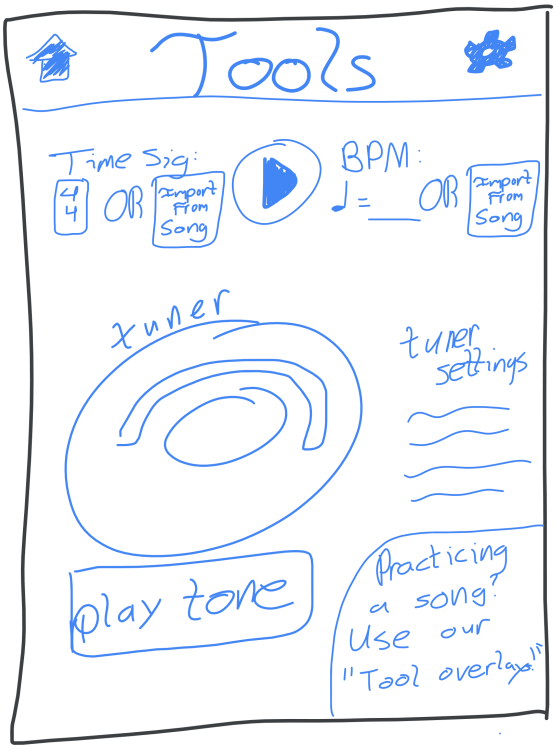




Original D: Jonathan Dempsey

Source submission, library and tools. Library, viewer and page indicator informed integration. Tools and unrelated controls remain outside the MVP.





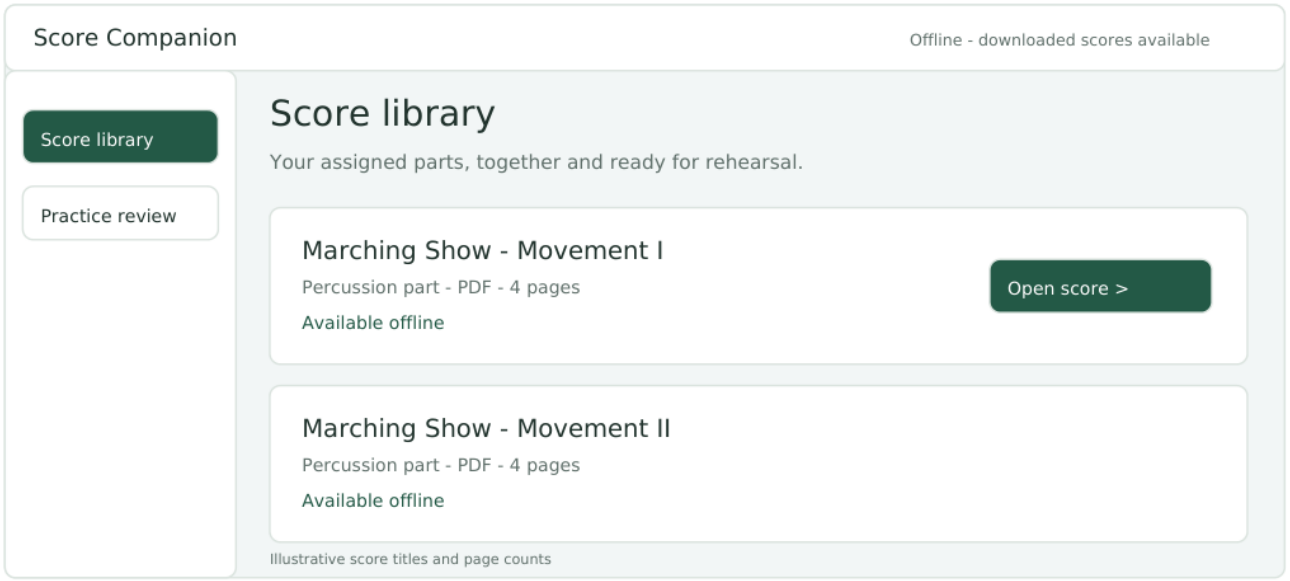
Original C: Ankit Joshi

Source submission, screen 1 of 5. Original attribution and evidence annotations preserved.

Screen 1 - Offline score library

P7 Rapid Prototyping | Responsive web app | Contributed by: C - Ankit Joshi

User action: Open the correct downloaded instrument part.



Design decision: Keep assigned parts together and show offline availability.

Interview evidence: “I download everything to where it's all in the same place in the offline section so I can easily bounce between.” — Interview 1

Segment: Format-Constrained Score Navigators

Next: open the score.

1 / 5

Static UI prototype. Sample score content; offline storage and PDF handling are not implemented.

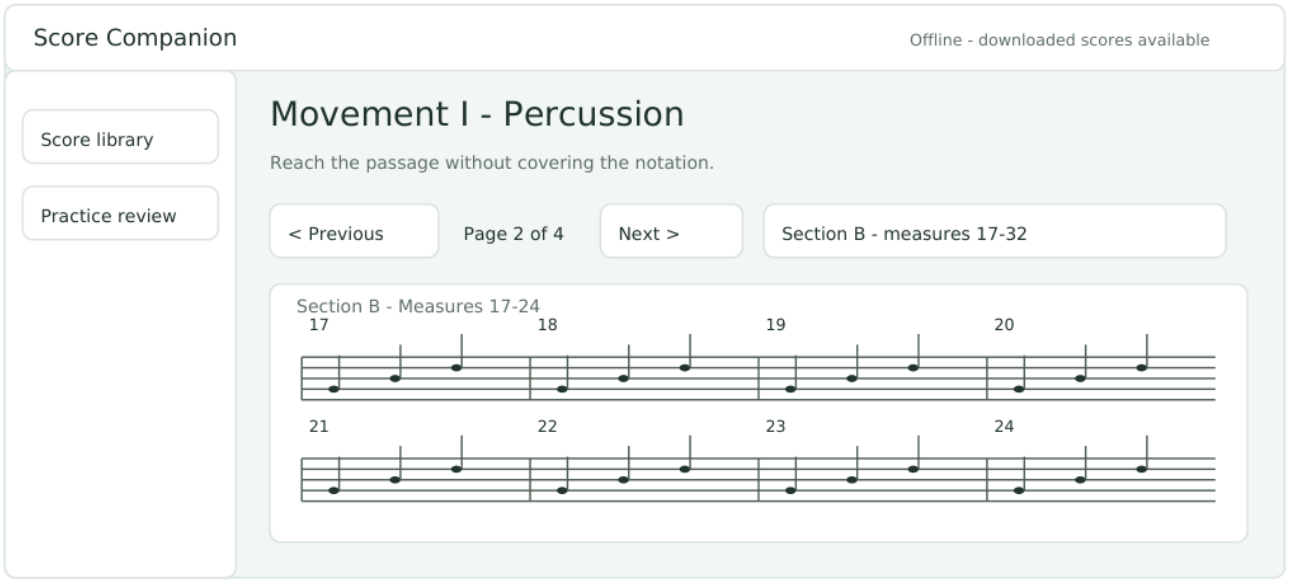
Original C: Ankit Joshi

Source submission, screen 2 of 5. Original attribution and evidence annotations preserved.

Screen 2 - Score viewer and navigation

P7 Rapid Prototyping | Responsive web app | Contributed by: C - Ankit Joshi

User action: Use page or section controls to reach the passage.



Design decision: Keep navigation outside the score so controls do not obscure notation.

Interview evidence: “MuseScore doesn't have a good zoom functionality to it and the UI gets in the way of the notes. No markup either for finger positions.” — Interview 4

Segment: Format-Constrained Score Navigators

Next: capture the correction at measure 24.

2 / 5

Static UI prototype. Sample score content; offline storage and PDF handling are not implemented.

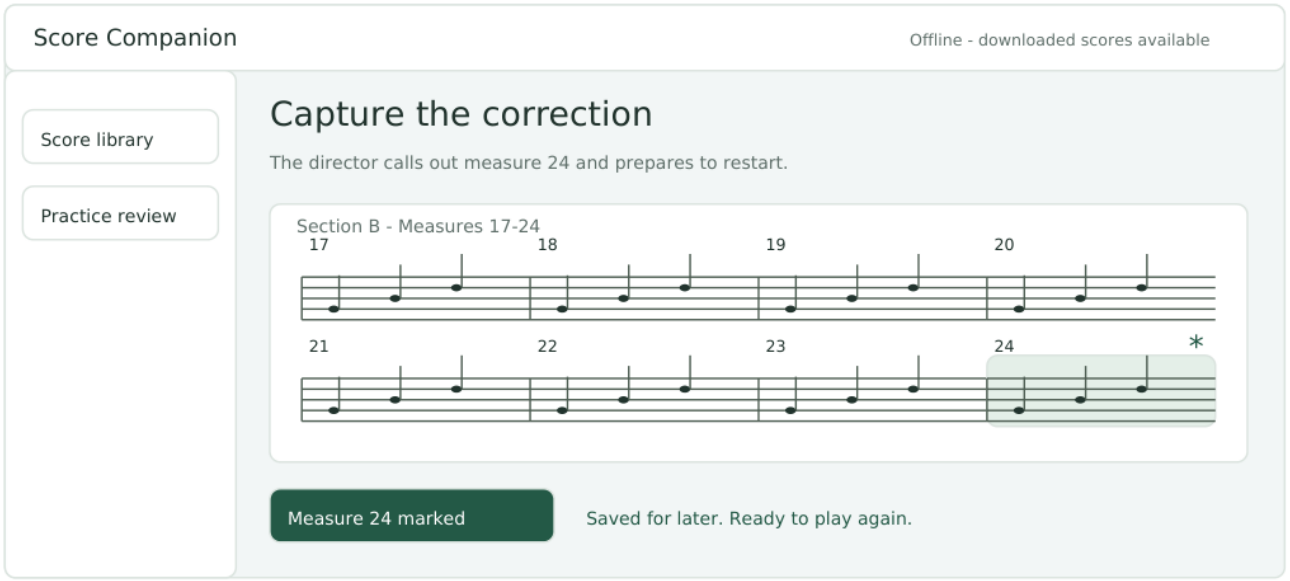
Original C: Ankit Joshi

Source submission, screen 3 of 5. Original attribution and evidence annotations preserved.

Screen 3 - One-action measure marker

P7 Rapid Prototyping | Responsive web app | Contributed by: C - Ankit Joshi

User action: Mark measure 24 and stay ready for the next entrance.



Design decision: Use one action to flag the identified measure without opening a typing dialog.

Interview evidence: “Often we don’t have enough time to pick up our pencils, write it down, and get ready to play again.” — Interview 7

Segment: Reactive Score Annotators

Next: add context after rehearsal.

3 / 5

Static UI prototype. Sample score content; offline storage and PDF handling are not implemented.

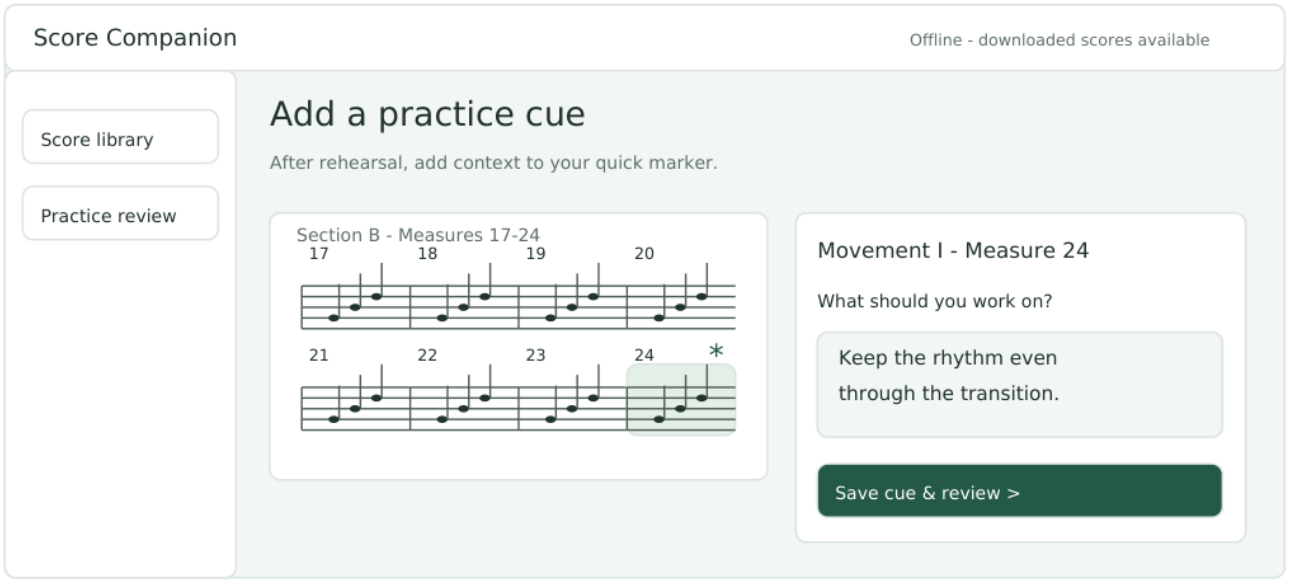
Original C: Ankit Joshi

Source submission, screen 4 of 5. Original attribution and evidence annotations preserved.

Screen 4 - Measure-linked practice cue

P7 Rapid Prototyping | Responsive web app | Contributed by: C - Ankit Joshi

User action: Add a concise cue after rehearsal and save it to the marked passage.



Design decision: Separate the immediate marker from later note entry; keep the measure context visible.

Interview evidence: “If for whatever reason we’re moving really quickly or doing a lot of sight reading, I star the measure. And when I look at it later by myself, I’ll see the star and write it in.” — Interview 7

Segment: Reactive Score Annotators

Next: save the cue and review.

4 / 5

Static UI prototype. Sample score content; offline storage and PDF handling are not implemented.

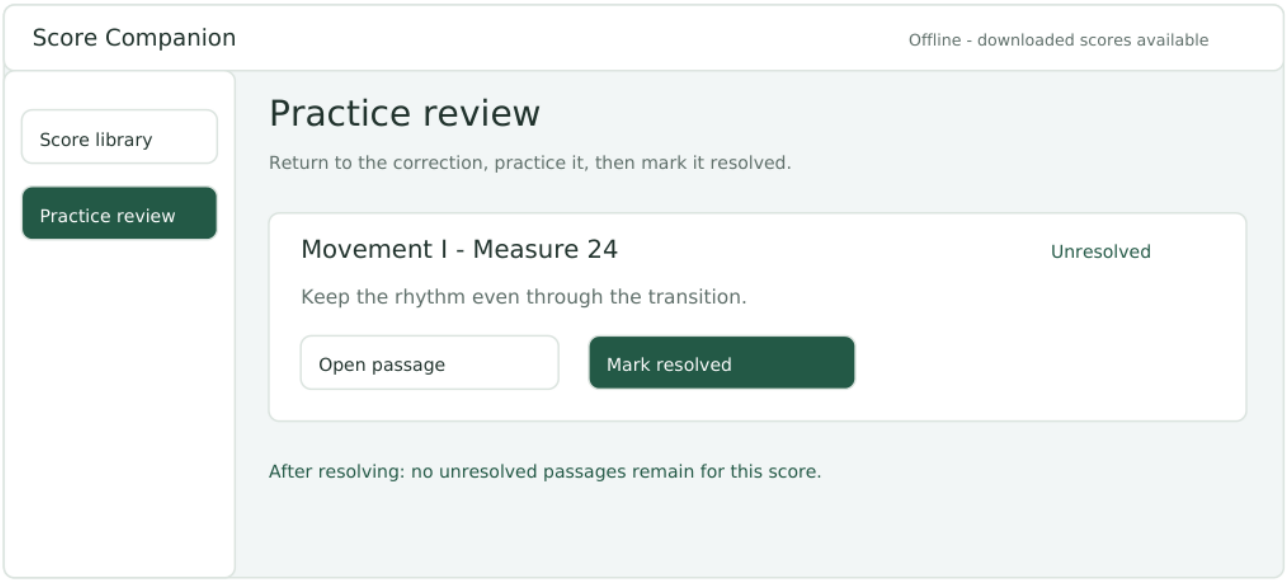
Original C: Ankit Joshi

Source submission, screen 5 of 5. Original attribution and evidence annotations preserved.

Screen 5 - Unresolved-passage review

P7 Rapid Prototyping | Responsive web app | Contributed by: C - Ankit Joshi

User action: Open the flagged passage, practice the correction, and mark it resolved.



Design decision: Keep unresolved corrections in a focused review list. Returning to and resolving digital markers remains an assumption to test.

Interview evidence: “Yeah. I remember it, and then I practice it the right way until it becomes muscle memory.” — Interview 1

Segment: Preparation-First Internalizers / Reactive Score Annotators

Completion: practice the passage and mark the correction resolved.

5 / 5

Static UI prototype. Sample score content; offline storage and PDF handling are not implemented.